User Manual

Single Photon SPAD Detector

SPAD-050-C-B-FC
SPAD-050-C-R-FC

User Manual – Version v1.1.0
04 March 2026

Contents

1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the Single Photon SPAD Detector. It forms an integral part of the product and must remain accessible to the personnel responsible for the device. If the device is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.

Before performing any operation, the entire manual must be read carefully and the relevant sections reviewed in detail.

1.1.1 User Qualification

The device is intended for use by qualified personnel familiar with laboratory electronic instrumentation and responsible for the installation, operation, and maintenance of the device throughout its service life. It is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the device only under supervision by qualified personnel.

1.1.2 Revision History

Document Revision: v1.1.0- 04 March 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period (10 years). During this period, documentation may be requested from the manufacturer.

1.2 Symbols and Conventions

The following graphical symbols and textual conventions may appear in this manual or on the device.

Provides important supplementary information.

Indicates procedures that, if not observed, may result in damage to the device or associated equipment.

Indicates procedures that, if not observed, may result in injury or serious damage.

(a) General hazard symbol **(b)** Mandatory reading of the manual

Figure 1.1: The device may include the pictograms shown above.

All safety symbols must remain clearly visible throughout the lifetime of the device.

1.3 Manufacturer Information

Manufacturer:	FLIM LABS S.r.l.
Address:	Via della Farnesina, 3 00135 Rome (RM), Italy
Telephone:	+39-329 761540
E-mail:	info@flimlabs.com
Website:	www.flimlabs.com

1.4 Declaration of Conformity

The device is supplied with a Declaration of Conformity drawn up in accordance with applicable European legislation. Before use, verify that the Declaration of Conformity is present. A copy of the Declaration of Conformity is shown in Document 1.1.

1.5 Product Label

The product label affixed to the device contains the required regulatory markings and identification information including:

- Product name and model
- Manufacturer identification
- Regulatory markings
- Electrical specifications

An example product label is shown in Figure 1.2. Disposal requirements associated with the WEEE marking are described in Section 7.2.

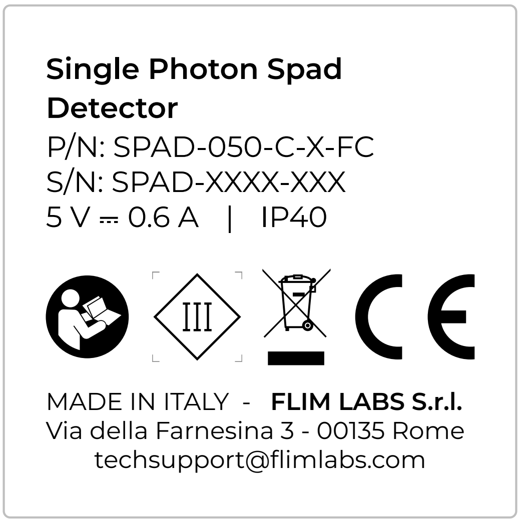

Figure 1.2: Product label

> Safety pictograms performing a protective function must not be removed, covered, or damaged. If the label becomes illegible or damaged, contact authorized service for replacement.

1.6 Warranty and Liability

1.6.1 Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

EU Declaration of Conformity

CE

Device Information	
Type:	Single Photon SPAD Detector
Model:	Single Photon SPAD Detector
Part Number:	SPAD-050-C-B-FC, SPAD-050-C-R-FC
Serial Number:	As indicated on the product label

Manufacturer Information	
Manufacturer:	FLIM LABS S.r.l.
Address:	Via della Farnesina, 3 00135 ROME (RM), Italy
Telephone:	+39 – 329 761540
E-mail:	info@flimlabs.com
Website:	www.flimlabs.com

The present declaration of conformity is issued under the sole responsibility of the manufacturer. The device complies with the following directives and regulations:

Applicable Directives and Regulations

- Directive 2014/35/EU, known as the "Low Voltage Directive"
- Directive 2014/30/EU, known as the "Electromagnetic Compatibility Directive"
- Directive 2011/65/EU, known as "RoHS"
- Delegated Directives (EU) 2015/863 and (EU) 2017/2102 amending Directive 2011/65/EU
- Directive 2012/19/EU, known as "WEEE"
- Regulation (EU) 2023/988 on General Product Safety

The manufacturer undertakes to provide, upon a duly reasoned request from the national authorities, all relevant information regarding the device.

Signed for and on behalf of FLIM LABS S.r.l.:

Place and Date	**Signature**
Rome, 24/10/2025	Alessandro Rossetta Alessandro Rossetta, CEO

Document 1.1: EU Declaration of Conformity

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the device is operated and maintained according to this manual. Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty. Full warranty terms are available in the manufacturer's General Terms and Conditions published on the official website.

To submit a warranty claim, provide the following:

- Product type/version
- Proof of purchase (including date)
- Detailed description of the issue

1.6.2 Limitation of Liability

The device has been designed, manufactured, and tested in accordance with applicable European directives and internal quality standards. The manufacturer is liable exclusively for defects attributable to manufacturing faults or non-conformity with declared specifications, within the limits defined in the warranty terms.

The manufacturer shall not be liable for damages resulting from improper use, including but not limited to:

- Use outside the intended purpose
- Failure to follow this manual or applicable safety instructions
- Installation or operation by unqualified personnel
- Operation outside specified electrical or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts
- Integration into systems not compliant with applicable safety regulations

The user is responsible for ensuring that the device is installed, operated, and integrated in accordance with this manual and all applicable local, national, and international regulations.

Except where otherwise required by mandatory law, the manufacturer shall not be liable for indirect, incidental, or consequential damages, including but not limited to loss of data, loss of profit, or system downtime.

2 Device Description

This section describes the functional characteristics, intended application, and technical features of the Single Photon SPAD Detector.

2.1 Functional Description

The Single Photon SPAD Detector (SPAD) is a compact single-photon detection module designed for the detection of extremely low-intensity optical signals down to the single-photon level. It is intended for applications in fluorescence lifetime imaging microscopy (FLIM), optical spectroscopy, photon counting, and time-resolved measurements in scientific and laboratory environments.

The device converts incident optical radiation into digital electrical pulses by means of an integrated SPAD (Single Photon Avalanche Diode) sensor. Incoming photons coupled via an optical fiber are detected by the SPAD operated in Geiger mode. Each detected photon event generates a fast electrical avalanche, which is internally processed and converted into a standardized digital output pulse. The resulting photon events are available at the SMA output in LVTTL format and via the USB Type-C interface in LVDS format for direct integration with data acquisition systems.

The product is available in two variants that differ exclusively in their spectral response characteristics:

- **SPAD-050-C-B-FC**: 370 nm to 900 nm, centered at 450 nm.
- **SPAD-050-C-R-FC**: 400 nm to 1000 nm, centered at 630 nm.

Both variants integrate a thermoelectric cooling (TEC) system to stabilize detector performance and reduce dark count rate.

The device can be powered via the USB Type-C interface or by an external 9 V DC power supply using a 2.1/5.5 mm coaxial DC connector.

The module features:

- Fiber-coupled optical input via multimode FC/PC connector
- Digital SMA output (2.5 V LVTTL, 50 Ω)
- USB Type-C interface with LVDS output
- Integrated thermoelectric cooling (TEC)
- Compact aluminum enclosure suitable for system integration
- Dual power supply options (USB Type-C or external 9 V DC)

2.2 Intended Use

The Single Photon SPAD Detector is intended as a compact single-photon detection module for converting incident optical radiation into digital photon event signals by professional users in research laboratories, universities, or industrial environments. It may be used as a stand-alone detector unit or integrated into custom optical and time-resolved measurement systems.

2.3 Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

2.4 Components Overview

Figure 2.1 shows an overview of the key components of the device. Those include:

1. 2.1/5.5 mm coaxial DC connector for power supply
2. LED status indicator
3. Digital SMA output (2.5 V LVTTL, 50 Ω)
4. 2x dual function USB Type-C ports: can be used as digital outputs (LVDS) or for power supply
5. Fiber-coupled optical input via FC/PC multimode connector

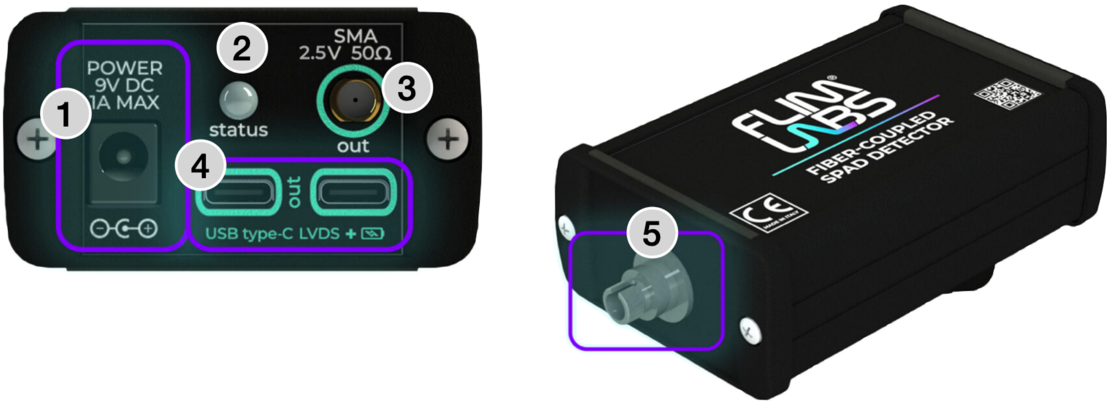

Figure 2.1: Key components of the Single Photon SPAD Detector.

2.5 Technical Specifications

The electrical, signal, and environmental limits referenced throughout this manual are summarized in this section. The spectral sensitivity curves can be found in Section 2.5.1.

2.5.1 Sensor Specifications

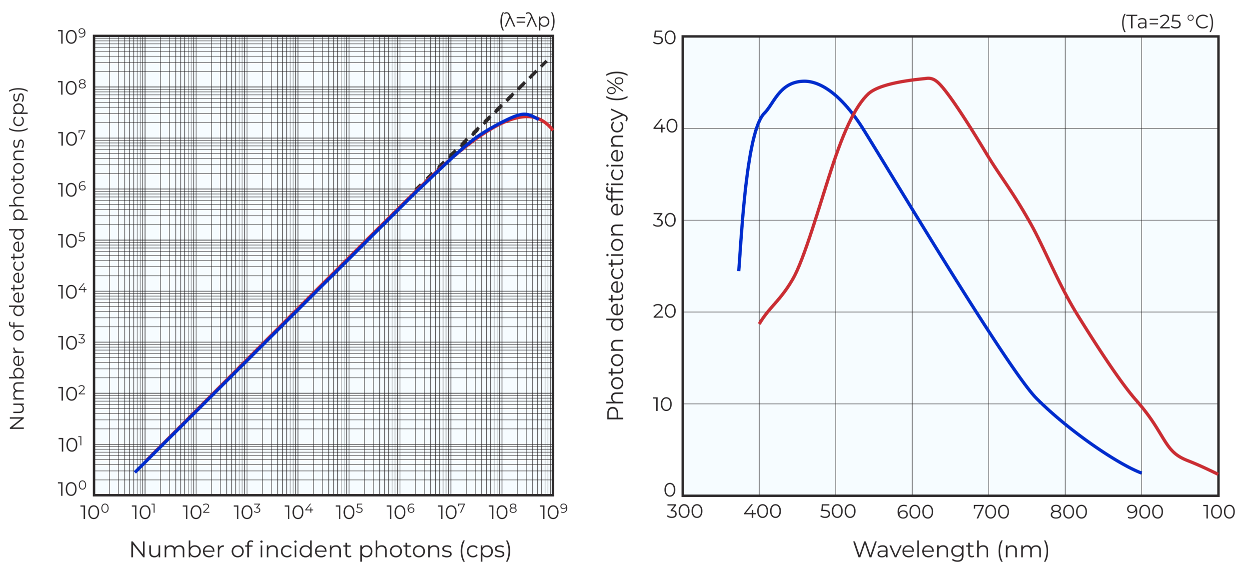

Figure 2.2: Detector photon efficiency and linearity curves over the spectral range of SPAD-050-C-B-FC **(blue)**, and SPAD-050-C-R-FC **(red)**.

Item	SPAD-050-C-B-FC	SPAD-050-C-R-FC
Electrical Specifications		
Electrical safety class	Class III	
Power supply	9 V DC (2.1/5.5 mm) or 5 V DC via USB Type-C	
Maximum current consumption	0.6 A	
Maximum power consumption	3 W	
Signal and Optical Specifications		
Sensor type	Single Photon Avalanche Diode (SPAD)	
Photosensitive area	50 µm	
Spectral response range	370 nm – 900 nm	400 nm – 1000 nm
Peak sensitivity	450 nm	630 nm
Maximum incident optical power	50 µW	
Maximum count rate	3.5 Mcps	3 Mcps
Dark count rate (typical)	7 cps	20 cps
Optical input	1x FC/PC multimode fiber connector	
Outputs	1x SMA (LVTTL) 1x USB Type-C (LVDS)	
Output pulse amplitude	SMA: 2.5 V LVTTL (50 Ω) USB Type-C: LVDS	
Output pulse width	10 ns	
Timing jitter	65 ps RMS	
Impedance	50 Ω	
Physical Specifications		
Dimensions (L x W x H)	127 mm x 77 mm x 40 mm	
Weight	344 g	
Protection rating	IP40	
Environmental Specifications		
Operating temperature	0°C to +40°C	
Storage temperature	-20°C to +70°C	
Relative humidity	Up to 80% non-condensing	

Table 2.1: Technical specifications of the Single Photon SPAD Detector.

2.6 Mechanical Drawings and Dimensions

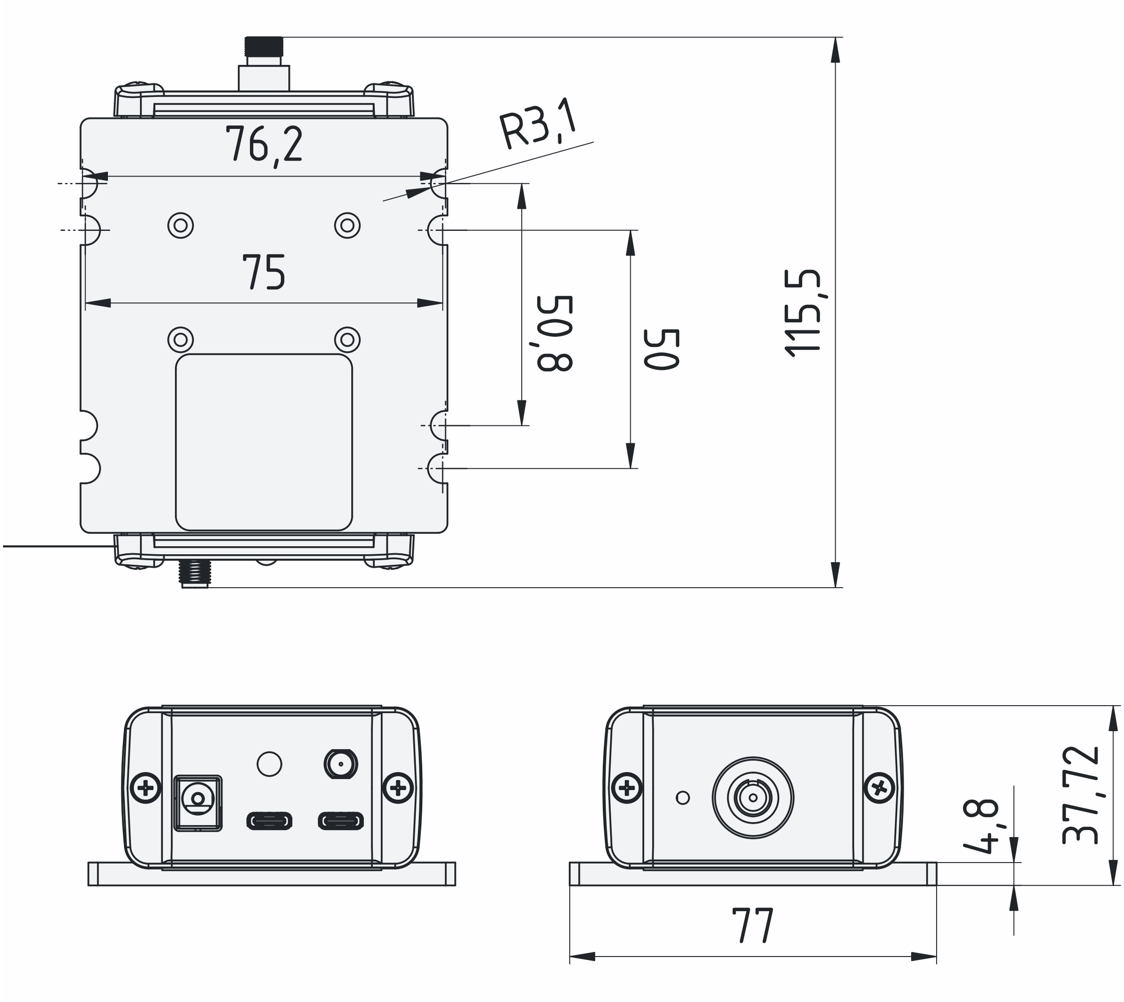

Figure 2.3: Technical drawing of the Single Photon SPAD Detector.

3 Safety Information

3.1 General Safety Instructions

Before installation and operation, ensure that this manual is complete and legible. If any part of the documentation is missing or unclear, contact the manufacturer before proceeding.

The device must be used only as described in this manual. Any operation not explicitly described in this manual is considered improper use.

This manual complements applicable national and local safety regulations. If local regulations impose stricter requirements, those regulations take precedence.

Despite the implemented safety measures, residual risks may remain if the device is not used as specified.

3.2 Electrical Safety

- Operate the device only with the specified power supply.
- Use undamaged, properly rated cables.
- Disconnect power before modifying signal connections.
- Do not open the enclosure. There are no user-serviceable internal components.
- Keep away from water sources and other liquids.
- Unauthorized repair or modification is not permitted.

3.3 Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.

3.4 Modification and Improper Use

The device must not be modified, altered, or repaired by unauthorized personnel. Use outside the intended application, operation beyond specified electrical limits, or failure to follow the procedures described in this manual may result in malfunction and void the warranty.

3.5 Documentation Retention

This manual must be kept accessible to personnel responsible for installation, operation, and maintenance. A replacement copy may be requested from the manufacturer if required.

4 Setup

This section describes the correct procedure for unpacking, installing, and configuring the device before operation.

4.1 Scope of Delivery

- SPAD detector
- Wall-mount DC power converter
- USB Type-C power cable
- SMA coaxial cable
- USB Type-C LVDS signal cable
- FC/PC multimode fiber patch cord
- Quick Start Guide
- Declaration of conformity

Verify that all items are present before proceeding with installation.

4.2 Unpacking and Inspection

The device is supplied in protective packaging to protect it during storage and transport. Handle the device carefully during unpacking to avoid mechanical damage.
After removing the packaging:

- Inspect the device and accessories for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that connectors and cables are intact.

If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.
Dispose of packaging materials responsibly and keep them out of reach of children.
The device is compact and lightweight and may be moved manually. When handling, support it securely to prevent accidental dropping.

4.3 Installation Requirements

Before installation and operation, ensure that the environmental conditions specified in Section 3.3 are met. Children must not have access to the installation or storage area.

Inspection of the FC/PC Optical Input

Perform a visual inspection to verify that the FC/PC optical port is free of dust, particles, or contamination that could obstruct or attenuate the optical signal transmitted through the fiber. If contamination is detected, clean the connector using appropriate fiber-optic cleaning tools before connection. Further details are specified in Section 6.1.

Verification of the Optical Input Signal

Before connecting any optical source to the detector input, verify that the maximum optical power of the light source does not exceed the maximum allowable incident optical power specified in Section 2.5. Exceeding the specified optical limits may permanently damage the sensor.

Never connect a laser source directly to the SPAD detector unless the optical power is verified to be within the specified input limits or appropriate attenuation is applied.

4.4 Mechanical Installation

Place the device on a stable laboratory surface with adequate space for cable routing and ventilation.

Ensure that:

- The device is positioned to prevent mechanical stress on connectors.
- Cables are routed safely and do not create tripping hazards.
- The power supply location is dry and away from water sources (e.g., sinks or splash zones).

Remove any protective dust or moisture covers before operation.

4.5 Electrical Connections

All electrical connections must be performed by qualified personnel who have read this manual. Connector locations are shown in Figure 2.1.
Before proceeding with the connections:

- Check the integrity of mechanical, electrical, and electronic components.
- Inspect all cables for visible damage.

Optical Connection

For operation of the SPAD detector, connect a multimode optical fiber with FC/PC connector to the optical input port of the device. Ensure proper mating of the connector and avoid excessive mechanical stress on the fiber.

Digital Output Connection

The digital output signal can be connected to a data acquisition system using one of the following interfaces:

- USB Type-C interface configured for LVDS output, using a compatible USB Type-C cable.
- Single-ended digital LVTTL output via the 50 Ω SMA connector, using a compatible SMA coaxial cable.

The SMA output allows integration with data acquisition systems from various manufacturers.

Power Supply Connection

The SPAD detector can be powered using one of the following methods:

- USB Type-C power input (5 V DC).
- External DC power supply via the 2.1/5.5 mm coaxial connector.

Recommended supply values are 9 V DC with a maximum current of 1 A, in accordance with SELV safety requirements.
During correct operation, the status LED emits green light.

4.6 Initial Configuration

The detector operates autonomously after completing all electrical and optical connections. If the device does not operate as expected after configuration, refer to the troubleshooting guidance provided in Section 5.4.

5 Operation

This section describes the functional behavior of the Single Photon SPAD Detector during use, including its operating principle and normal operating conditions.

Do not operate the device if it is damaged or shows signs of unsafe condition.

5.1 Principle of Operation

The SPAD detects incident photons coupled through an optical fiber and converts each detected photon event into a digital electrical pulse.

Incoming photons enter the device via the FC/PC optical connector and are directed onto an integrated Single Photon Avalanche Diode (SPAD) sensor operated in Geiger mode. When a photon is absorbed within the active area of the sensor, it triggers a controlled avalanche process, generating a fast electrical pulse. The internal electronics process this event and produce a standardized digital output pulse.

Each detected photon event results in:

- A 2.5 V LVTTL pulse at the SMA output (50 Ω)
- A differential LVDS signal at the USB Type-C output

The detector integrates thermoelectric cooling (TEC) to stabilize operating conditions and reduce dark count rate. The device does not perform optical amplification or signal conditioning beyond photon detection and digital pulse generation.

5.2 Normal Operation

Once the power supply and all signal connections have been correctly established, the detector operates autonomously. Normal operation is indicated by a green status LED.

The detector generates one digital pulse for each detected photon event. The resulting count rate depends on the incident photon flux, the spectral characteristics of the selected detector variant, and the overall system integration conditions.

5.3 Parameter Adjustment

The SPAD does not require user-adjustable internal parameters for standard operation. Performance characteristics such as count rate and signal stability depend primarily on:

- Optical alignment and coupling efficiency
- Incident optical power level
- Environmental temperature
- Compatibility with the connected data acquisition system

5.4 Troubleshooting

Common faults and corrective actions are summarized in Table 5.1.

Table 5.1: Overview of common problems and recommended corrective actions.

Problem	Possible Cause	Recommended Action
Power and Device Status		
Device does not power on	Insufficient or incorrect power supply	Verify USB Type-C or external 9 V DC supply. Ensure the power source meets the specified voltage and current requirements.
Device does not power on	USB port does not provide sufficient current	Use a USB port capable of delivering adequate current (e.g., USB 3.0 port) or use the external 9 V power supply.
Device does not power on	Defective power cable	Replace the USB Type-C or power supply cable and verify proper connector seating.
Status LED not illuminated	Power connection not properly established	Check cable integrity and connector seating. Confirm that the power source is active.
Status LED not illuminated	Device still in startup phase	Wait a few seconds for the device initialization to complete and verify that the status LED turns green.
Optical Input		
No digital output signal	Optical fiber not connected or improperly connected	Verify correct FC/PC fiber connection and proper connector mating.
No digital output signal	No optical signal present at detector input	Verify that the light source is active and that the optical path is correctly aligned.
No digital output signal	Optical port blocked by dust or contamination	Inspect the FC/PC connector visually and clean the optical port and fiber connector using appropriate fiber-optic cleaning tools.
No digital output signal	Incident optical power below detection threshold	Verify correct optical alignment and sufficient signal intensity. Confirm compatibility with the selected detector variant.
Signal Output and System Integration		
No digital output detected by acquisition system	Incorrect output interface selection (SMA / LVTTL or USB Type-C / LVDS)	Confirm that the correct output port is used and that the acquisition system is configured for the selected signal standard.
Output signal unstable or irregular	Mechanical stress on fiber or connectors	Ensure fiber and signal cables are properly secured and not under tension.
Detector Performance		
Excessive dark counts	High ambient temperature or strong background light	Operate within specified environmental limits and shield the detector from stray light.

Problem	Possible Cause	Recommended Action
Output count rate lower than expected	Acquisition system dead time or saturation	Verify compatibility of acquisition system settings and ensure system bandwidth is sufficient.

If any unexpected or potentially hazardous anomaly occurs during startup or operation, switch off the device immediately and disconnect it from the power source. Contact technical support before further use.

6 Maintenance and Service

This section describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the device. All maintenance activities must be performed by qualified personnel who have read and understood this manual.

> Before performing any cleaning or physical maintenance operation, the device must be switched off and disconnected from the power source.

Maintenance must be carried out in compliance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.

6.1 Cleaning

The device requires periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from the exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors or internal components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas (e.g., USB ports), a soft-bristled brush or compressed air may be used.

Optical Port and Connector Cleaning

Proper cleaning of optical connectors is essential to ensure reliable optical coupling and to prevent signal attenuation or permanent damage to the optical interface. Always inspect the ferrule end faces of fiber patch cables and the FC/PC optical bulkhead before connecting them to the device.

Ferrule end faces may be cleaned using commercially available lint-free fiber cleaning wipes or by applying a small amount of isopropanol (IPA) to a clean lint-free cloth.

The FC/PC optical bulkhead is best cleaned using a dedicated fiber-optic cleaning pen or cleaning stick specifically designed for bulkhead connectors. Follow the instructions provided by the manufacturer of the cleaning tool.

6.2 Preventive Maintenance

The device does not require routine internal maintenance. Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear or deformation

If any abnormal condition is detected, discontinue use and contact technical support.

6.3 Service and Repairs

The device contains no user-serviceable internal components. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the device.
- Do not attempt internal repair.
- Contact the manufacturer's technical service for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts. Further warranty conditions are described in Section 1.6.1.

> Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.

7 Decommissioning and Disposal

This section describes procedures for removing the device from service and disposing of it in accordance with applicable regulations.

7.1 Decommissioning Procedure

The device is designed for long-term laboratory use. When it reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the device from all power sources.
- Removing all signal connections.
- Ensuring that the device cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

7.2 Disposal

The device must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment. In accordance with Directive 2012/19/EU (WEEE), the device must not be disposed of with household waste. The product label includes the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment (see Section 1.5). Contact your local waste management authority or certified collection center for proper disposal procedures.

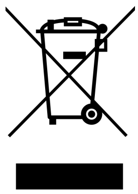

Figure 7.1: WEEE symbol